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To the Editors of *Remote Sensing* (MDPI)

Dear Editors,

We are pleased to submit our manuscript, "Predicting Tidal-Sampling Bias of Sun-Synchronous Satellites from Overpass Phase: Theory and Validation on Macrotidal Coasts," for consideration as a research article in *Remote Sensing*.

The gap we address. The waterline method underpins most operational intertidal Digital Elevation Models (DEMs), including the Digital Earth Australia Intertidal product (Bishop-Taylor et al., 2019), the global tidal-flat distributions of Murray et al. (2019, *Nature*), and a recent multi-sensor optimisation for the Korean west coast by Lee et al. (2025, *Estuarine, Coastal and Shelf Science* 318, 109235). All of these rely on the assumption that satellite imagery samples the local tidal cycle without systematic bias. Bishop-Taylor et al. (2019) flagged this assumption as questionable for sun-synchronous orbits, but no closed-form prediction of the resulting bias has been published, and the empirical workarounds proposed since — such as Lee et al.'s 5-month multi-sensor data-collection window — leave the *physical reason* for the optimum unquantified.

What we contribute. Using 5,082 cloud-screened Landsat-8/9 and Sentinel-2 scenes (2020–2024) over five tidal flats spanning the entire western and southern Korean coast, coupled with hourly Korea Hydrographic and Oceanographic Agency (KHOA) gauge observations, we derive and validate a single closed-form bias-prediction model: mean bias $\approx \beta \cdot \mathrm{A} \cdot \langle \cos \theta \rangle$, where A is the local tidal amplitude and $\langle \cos \theta \rangle$ the mean cosine of the satellite-overpass phase. The model explains 98 % of the cross-site, cross-sensor variance ($\beta = 1.78$, 95 % CI 1.44–1.91), is stable across years, seasons, and sensors (leave-one-site-out RMSE = 0.16 m, 15/15 correct sign), and translates via quantile mapping into a DEM-equivalent vertical RMSE of 0.36–1.09 m — equivalent to permanently unsampled intertidal bands up to 2.5 km wide at Ganghwa-do. The departure $\beta > 1$ is itself a *physical diagnostic*: it decomposes analytically into a spring–neap × overpass-phase covariance term rather than residual scatter. Crucially, replacing the local KHOA reference with the global FES2022b ocean tide model reproduces the fit ($\beta = 1.70$, $R^2 = 0.983$, LOO RMSE = 0.11 m) with the bias sign correct at every site, confirming that the correction is applicable worldwide without any local tide-gauge access. To our knowledge this is the first analytical, *a priori* bias-correction tool of this kind for the optical-waterline workflow. We also report the first quantitative demonstration that the bias *changes sign* across a regional amphidromic phase gradient — negative (low-tide) on the macrotidal west coast, positive (high-tide) at south-coast Suncheon Bay.

Relation to recent Korean tidal-flat DEM work. We note two recent and highly relevant Korean studies in this topical neighbourhood: (i) Lee et al. (2025, ECSS 318, 109235) empirically optimised an optical+SAR fusion DEM on the Taean Peninsula, reporting a 5-month minimum data-collection window with a mean absolute error of 25.6 cm against UAV-LiDAR; and (ii) Lee, K. et al. (2022, *Fron-*

tiers in Marine Science 9, 810549) showed that altimetry-sampling-phase shifts inflate apparent sea-level-rise trends near Incheon. Our paper is complementary to, not redundant with, both: Lee et al. (2025) characterise the phenomenon *empirically* and propose an operational solution, whereas we provide the *first-principles model* that explains *why* their 5-month optimum exists and *predicts* its coast-dependent variation (Section 5.3); Lee, K. et al. (2022) address the *temporal* drift of sampling phase within altimetry footprints, whereas we address the *spatial* counterpart for the optical-waterline workflow. Geographically, our five sites *bracket without duplicating* the Taean Peninsula coverage of Lee et al. (Garorim Bay lies ~30 km north; Suncheon Bay extends the analysis to a south-coast amphidromic regime outside their study area). Section 1.3 and Table 1 of the manuscript place this paper explicitly in that landscape.

Why *Remote Sensing*. The correction tool is region-agnostic: it depends only on local tidal amplitude and satellite overpass phase, both available from any global tide model (FES2022b, TPXO9) and satellite ephemeris. The bipolar bias polarity we report is a *spatial* manifestation of a global mechanism that will recur wherever amphidromic phase gradients comparable to 1/4 of an M_2 cycle exist (Bay of Fundy, southern North Sea, the Wash, the Australian north coast). As a concrete mission-design implication, our model predicts that a two-satellite optical constellation with overpass times staggered by 3 h (1/4 M_2 cycle) would halve the systematic bias — a gain unattainable by any increase in revisit frequency on a single sun-synchronous orbit. The manuscript therefore (a) provides a ready-to-use *a priori* bias correction for the existing global waterline-DEM community, (b) supplies the physical basis for future multi-sensor DEM optimisation in macrotidal regions, and (c) informs the design of next-generation intertidal-observing missions — topics squarely within the scope of *Remote Sensing*'s coverage of satellite remote sensing of coastal and intertidal environments.

Submission status. The manuscript is original, has not been published or submitted elsewhere, and all authors have approved the submission. All raw KHOA observations and Google Earth Engine scene-metadata queries are publicly accessible; derived tables and analysis scripts will be deposited in a Zenodo-archived GitHub repository upon acceptance. The authors declare no conflicts of interest.

We hope you find the manuscript suitable for *Remote Sensing* and look forward to the review process.

Sincerely,

Taeyoon Song (on behalf of both authors)

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Suggested reviewers (international experts with no prior collaboration with the authors; e-mail addresses supplied in the submission system):

- Robbi Bishop-Taylor — Geoscience Australia (Digital Earth Australia Intertidal; tidal-aliasing diagnostics — the work this study extends)
- Stephen Sagar — Geoscience Australia (continental-scale intertidal extent and topography)
- Nicholas J. Murray — James Cook University, Australia (global tidal-flat distribution and change)
- Kuo-Hsin Tseng — National Central University, Taiwan (optical reconstruction of time-varying tidal-flat topography)

- Nan Xu — Shenzhen University, China (ICESat-2 + Sentinel-2 intertidal topography and vertical referencing)

Opposed reviewers: none.